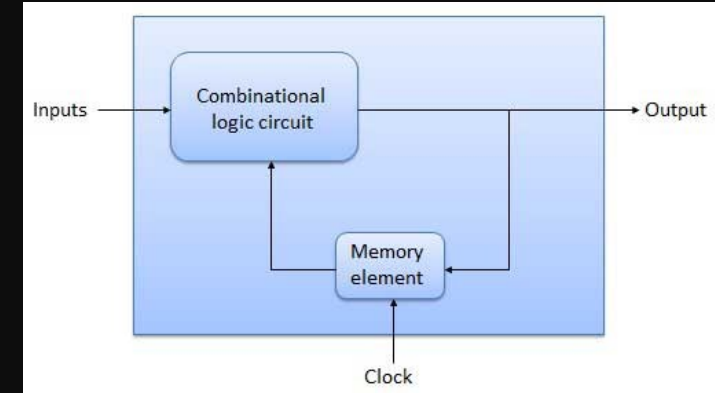


Lecture 6

Topics: Memory, Flip-flops and Sequential circuit



Probir Roy
Assistant professor,
CIS

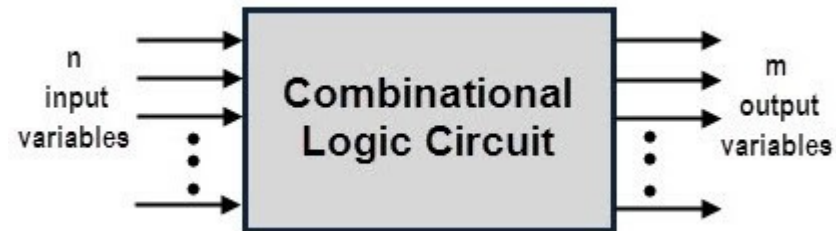
CIS-310: Computer Organization & Assembly Language

MW



Combinational Logic Applications

- Full adder
- Seven-Segment Displays
- Decoders
- Multiplexers
- Demultiplexers



Memory cell (Latch/ flip-flop)

https://faculty.etsu.edu/tarnoff/ntes2150/Ch10_v02.pdf

Memory cell

- Computer memory is made from arrays of cells, each of which is capable of storing a single bit, a one or a zero. The goal is to send a logic one or a logic zero to a device, then after leaving it unattended for a period of time, come back and find the value still there.

Primitive Feedback Circuit using Inverters

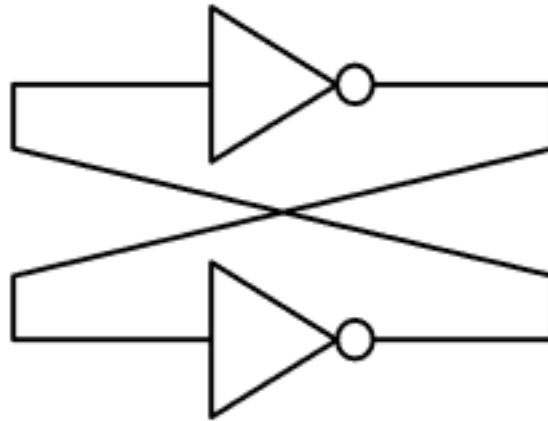


Figure 10-3 Primitive Feedback Circuit using Inverters

The problem with the circuit is that there is no way to modify the value that is stored

Primitive Feedback Circuit with NAND Gates

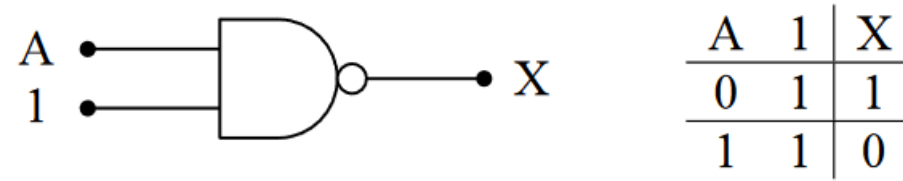


Figure 10-4 Operation of a NAND Gate with One Input Tied High

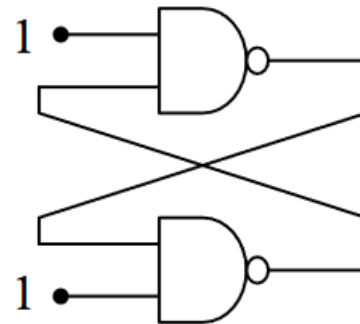


Figure 10-5 Primitive Feedback Circuit Redrawn with NAND Gates

Two Possible States of Circuit

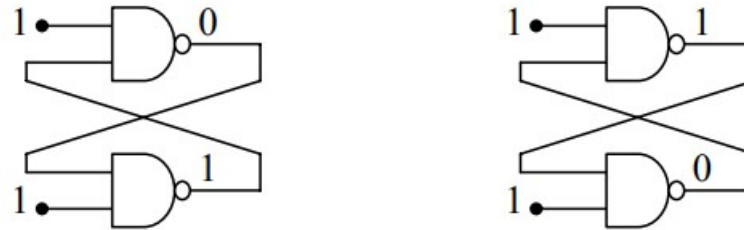
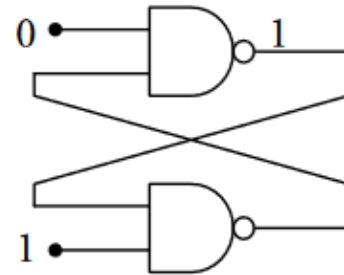
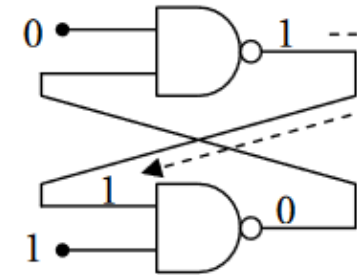


Figure 10-6 Only Two Possible States of Circuit in Figure 10-5

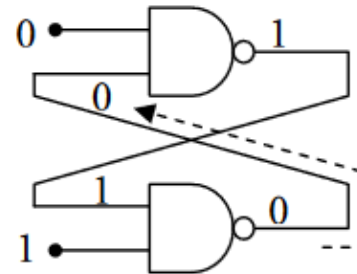
Operation of a Simple Memory Cell



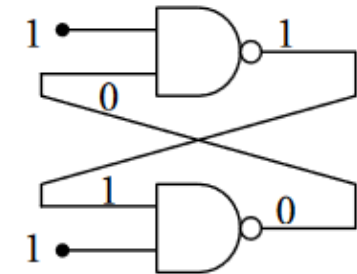
- a.) A zero to the free input of the top NAND gate forces a one to its output



- b.) That one passes to the bottom NAND which in turn outputs a zero



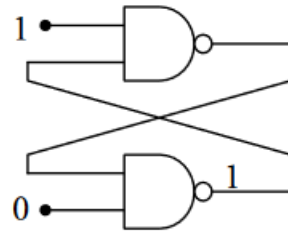
- c.) A zero from the bottom NAND returns to the lower input of the top NAND



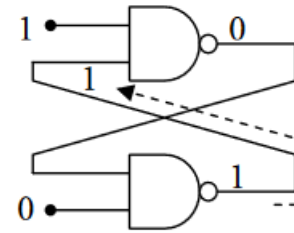
- d.) The second zero at the top NAND holds its output even if the free input returns to 1

Figure 10-7 Operation of a Simple Memory Cell

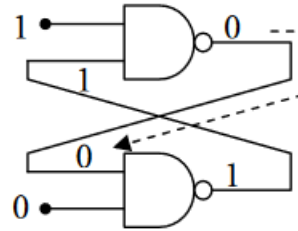
Operation of a Simple Memory Cell (continued)



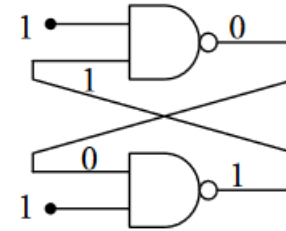
- a.) A zero to the free input of the bottom NAND gate forces a one to its output



- b.) That one passes to the top NAND which in turn outputs a zero

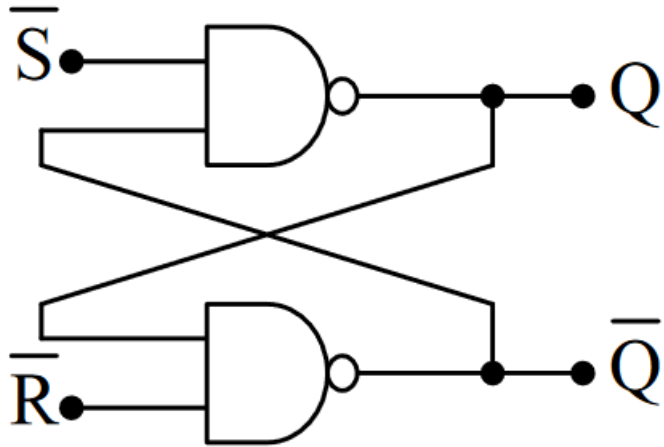


- c.) A zero from the top NAND returns to the lower input of the bottom NAND



- d.) The second zero at the bottom NAND holds its output even if the free input returns to 1

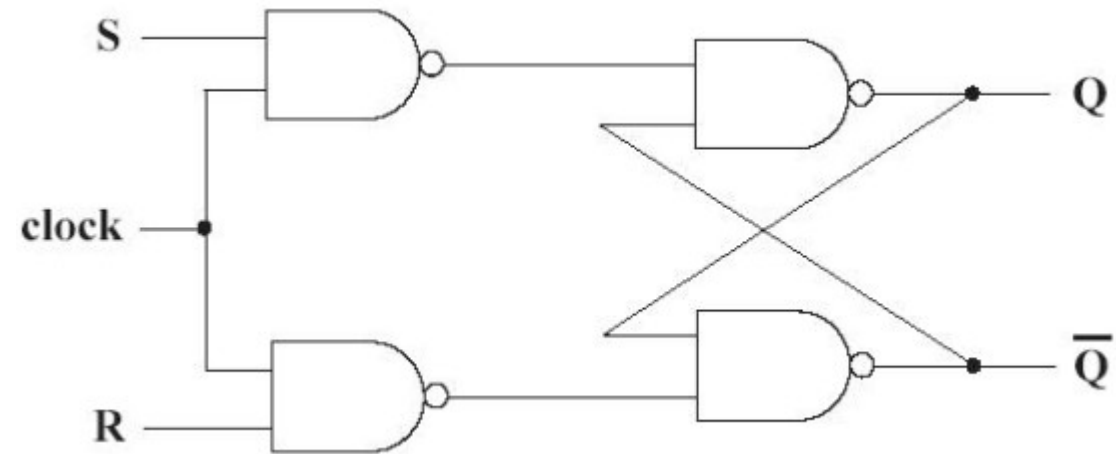
S-R Latch



$\overline{S}$	$\overline{R}$	Q	$\overline{Q}$
0	0	U	U
0	1	1	0
1	0	0	1
1	1	Q_0	$\overline{Q}_0$

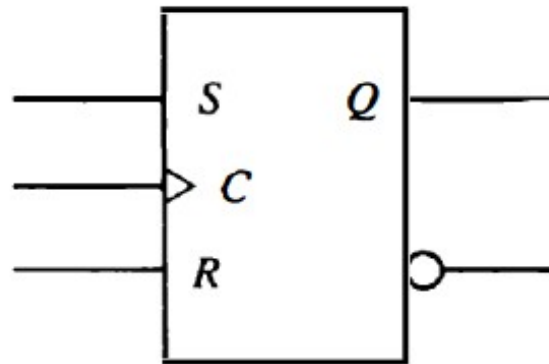
Figure 10-10 S-R Latch Truth Table

S-R flipflop



Memory cell

- SR flip-flop



(a) Graphic symbol

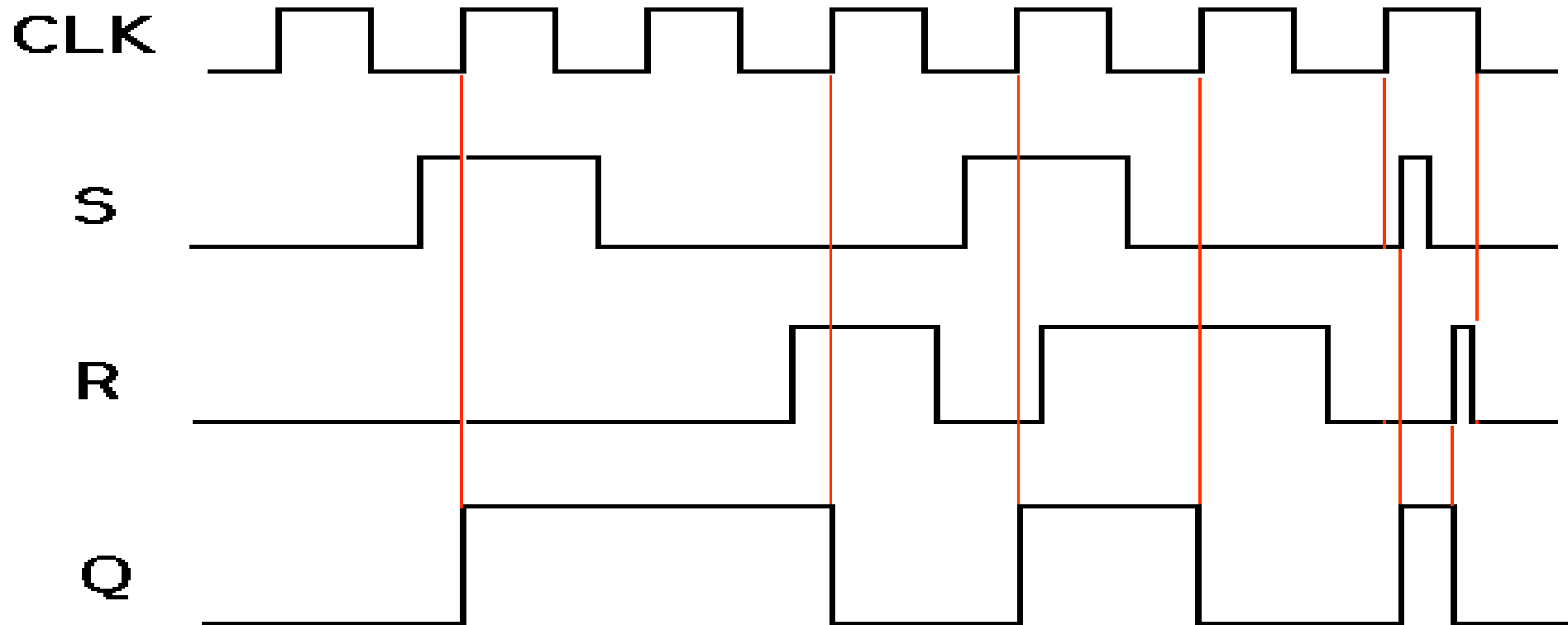
S	R	$Q(t+1)$	
0	0	$Q(t)$	No change
0	1	0	Clear to 0
1	0	1	Set to 1
1	1	?	Indeterminate

(b) Characteristic table

Clock pulse

- Synchronous sequential circuits employ signals that affect the storage elements only at discrete instants of time. Synchronization is achieved by a timing device called a clock pulse generator that produces a periodic train of clock pulses

SR flipflop clock signal

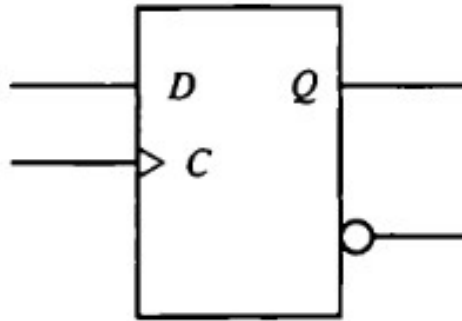


Edge-Triggered Flip-Flops

More flipflops

- D-flipflop
- JK-flipflop
- T-flipflop
- ...

D Flipflop



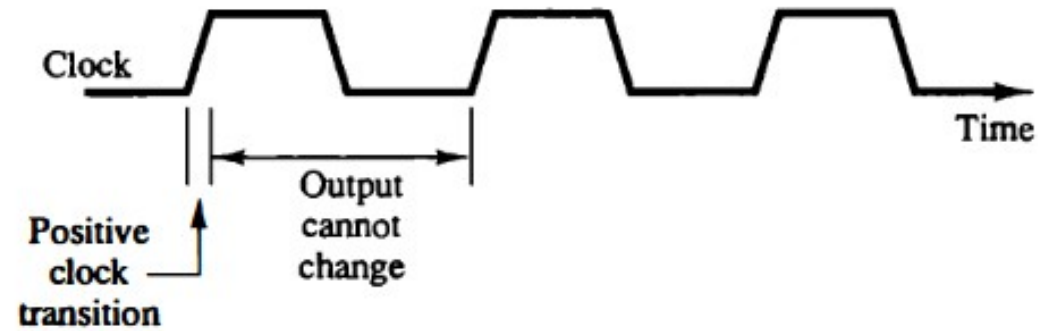
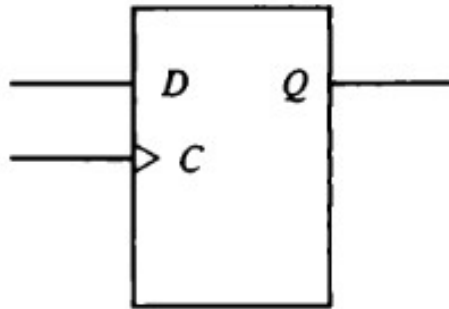
(a) Graphic symbol

D	$Q(t + 1)$
0	0
1	1

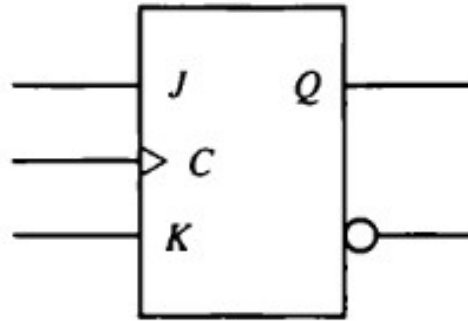
Clear to 0
Set to 1

(b) Characteristic table

Positive-edge-triggered D flip-flop



JK-Flipflop

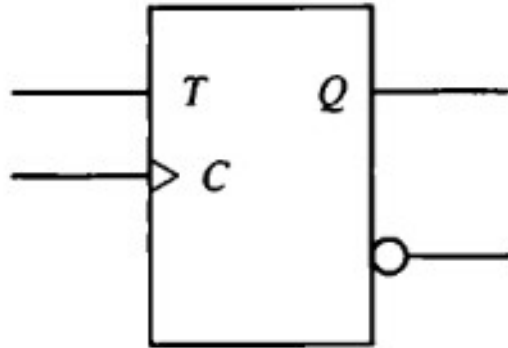


(a) Graphic symbol

J	K	$Q(t+1)$	
0	0	$Q(t)$	No change
0	1	0	Clear to 0
1	0	1	Set to 1
1	1	$Q'(t)$	Complement

(b) Characteristic table

T Flip-flop



(a) Graphic symbol

T	$Q(t+1)$	
0	$Q(t)$	No change
1	$Q'(t)$	Complement

(b) Characteristic table

Excitation Tables

TABLE 1-3 Excitation Table for Four Flip-Flops

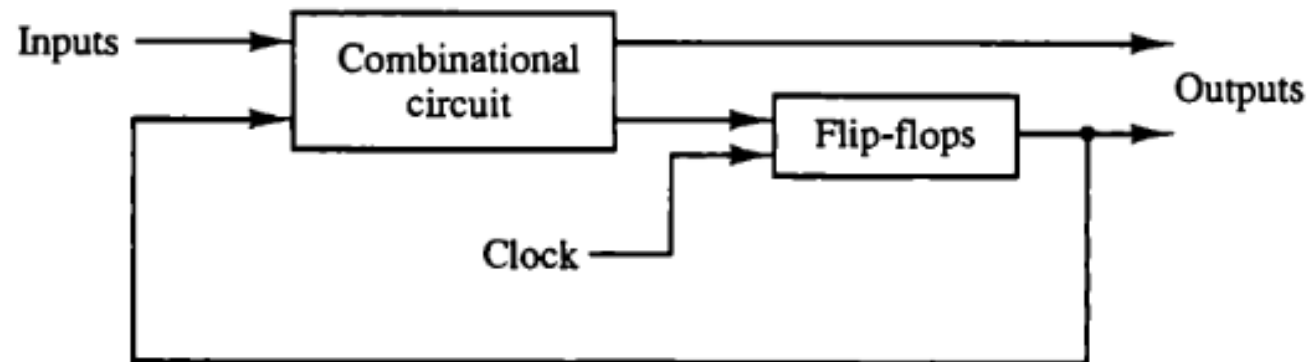
SR flip-flop				D flip-flop		
$Q(t)$	$Q(t + 1)$	S	R	$Q(t)$	$Q(t + 1)$	D
0	0	0	×	0	0	0
0	1	1	0	0	1	1
1	0	0	1	1	0	0
1	1	×	0	1	1	1

JK flip-flop				T flip-flop		
$Q(t)$	$Q(t + 1)$	J	K	$Q(t)$	$Q(t + 1)$	T
0	0	0	×	0	0	0
0	1	1	×	0	1	1
1	0	×	1	1	0	1
1	1	×	0	1	1	0

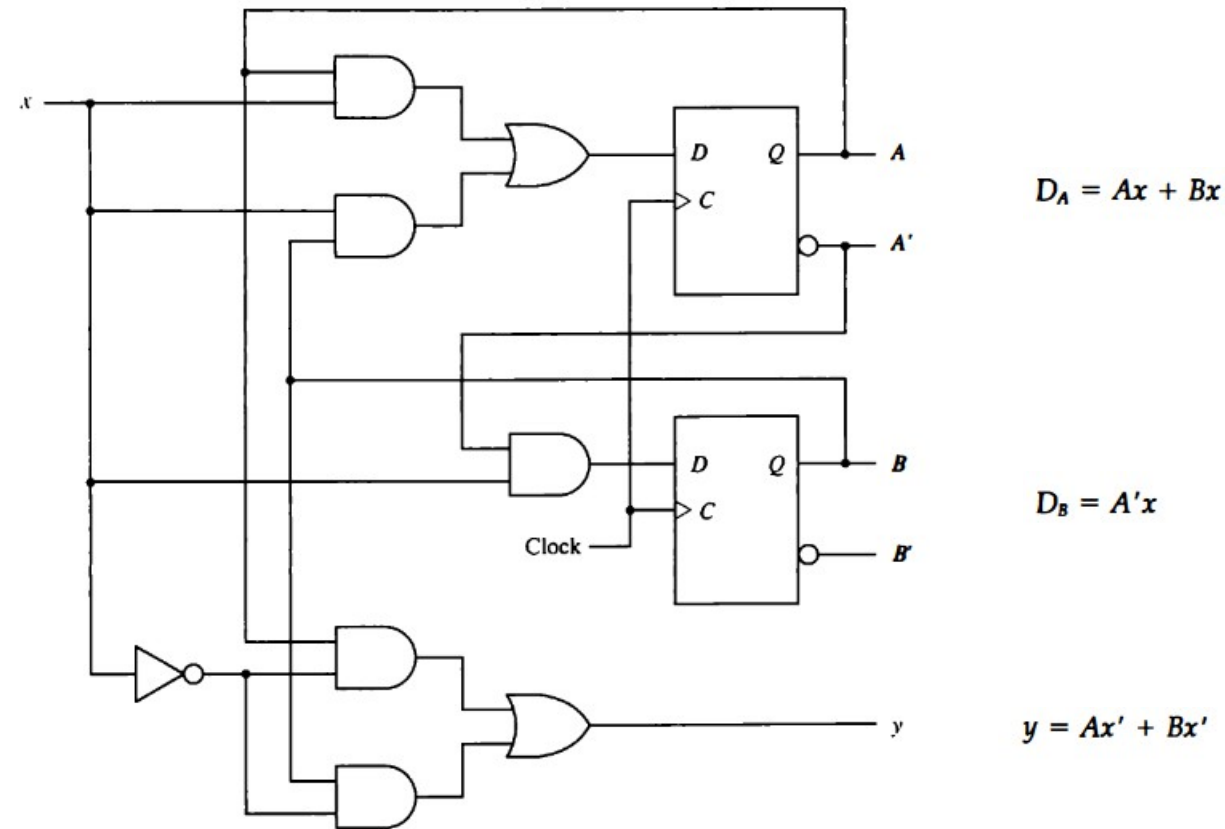
Sequential Logic Circuit

Sequential Logic Circuit

- A sequential circuit is an interconnection of flip-flops and gates. The gates by themselves constitute a combinational circuit, but when included with the flip-flops, the overall circuit is classified as a sequential circuit.



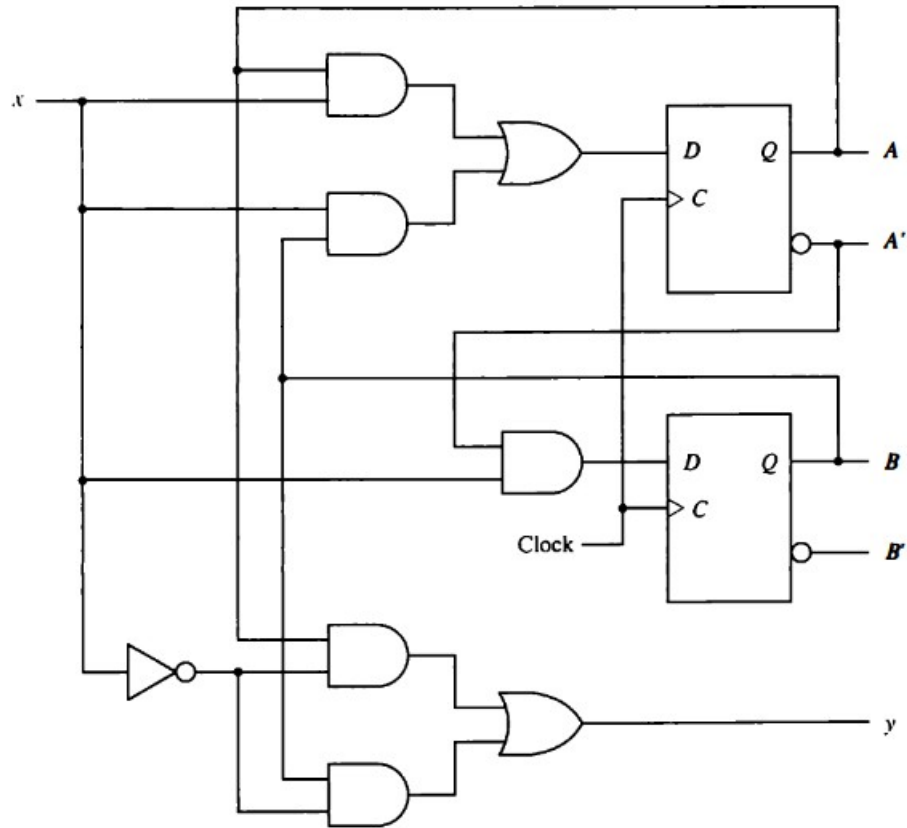
An example sequential circuit



State table

- The behavior of a sequential circuit is determined from the inputs, the outputs, and the state of its flip-flops.
- Both the outputs and the next state are a function of the inputs and the present state.
- sequential circuit is specified by a **state table** that relates outputs and next states as a function of inputs and present states.

State table (Cont)



$$D_A = Ax + Bx$$

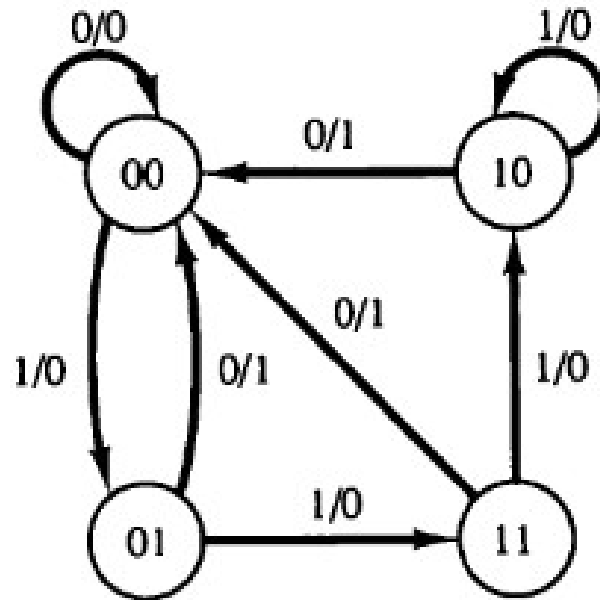
$$D_B = A'x$$

$$y = Ax' + Bx'$$

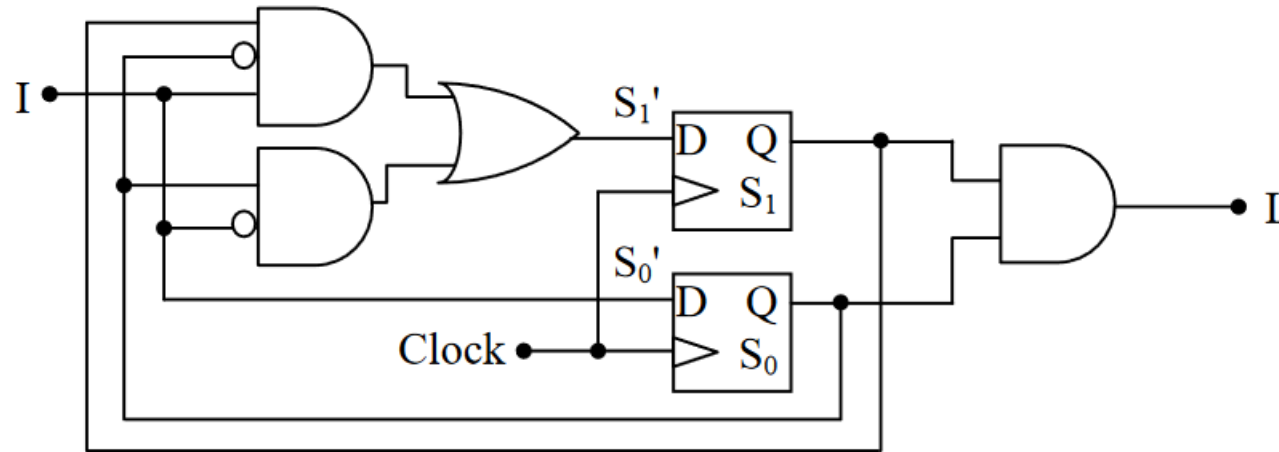
Present state		Input	Next state		Output
A	B		A	B	
0	0	0	0	0	0
0	0	1	0	1	0
0	1	0	0	0	1
0	1	1	1	1	0
1	0	0	0	0	1
1	0	1	1	0	0
1	1	0	0	0	1
1	1	1	1	0	0

State Diagram

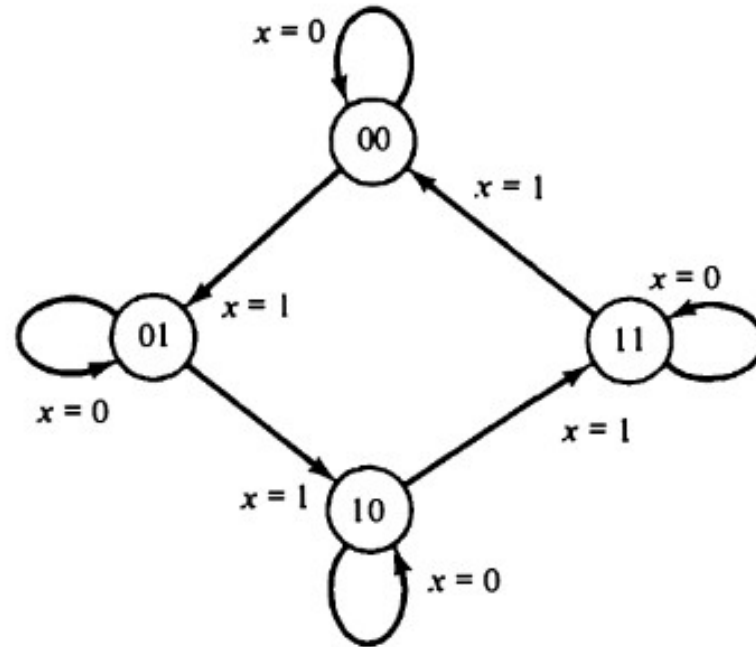
- The information available in a state table can be represented graphically in a state diagram.



QA: Derive the state diagram of the sequential circuit



Given a state-diagram, design sequential circuit

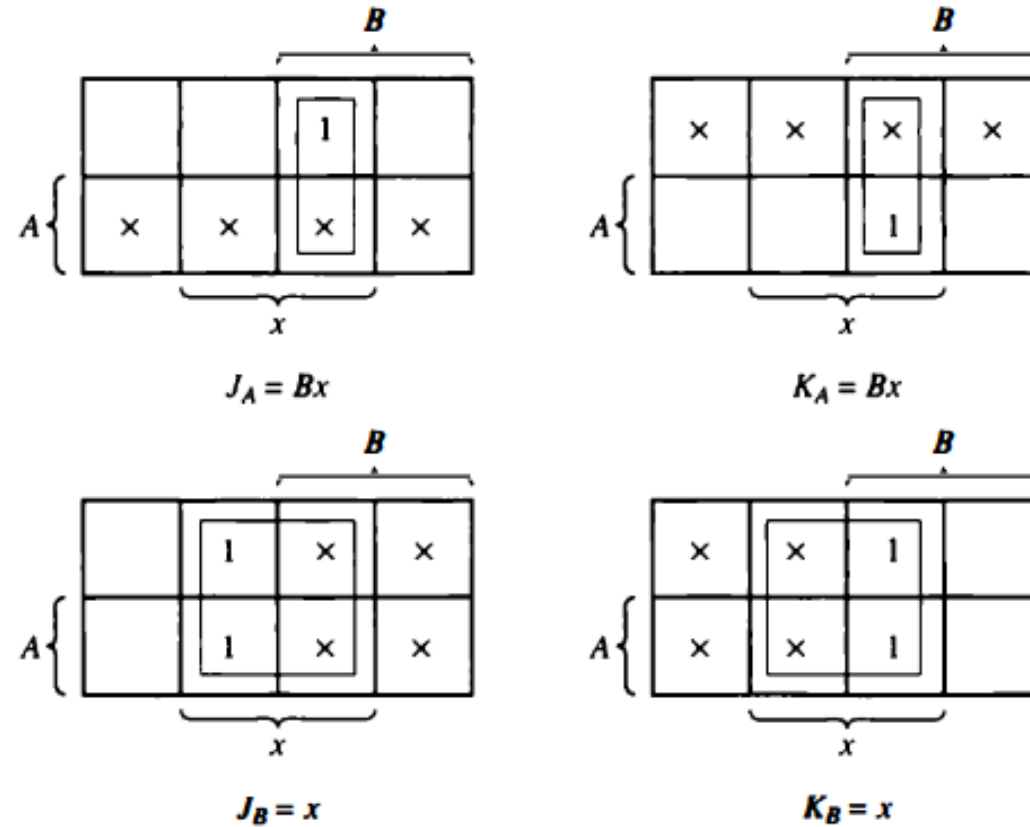


Implement using JK flip-flop

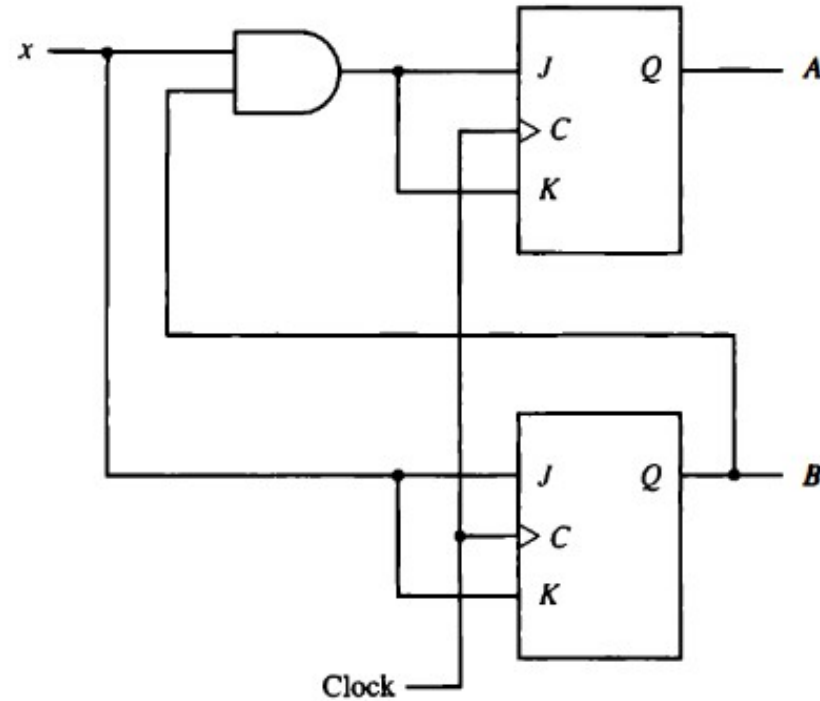
- excitation table

Present state		Input	Next state		Flip-flop inputs			
A	B		A	B	J_A	K_A	J_B	K_B
0	0	0	0	0	0	×	0	×
0	0	1	0	1	0	×	1	×
0	1	0	0	1	0	×	×	0
0	1	1	1	0	1	×	×	1
1	0	0	1	0	×	0	0	×
1	0	1	1	1	×	0	1	×
1	1	0	1	1	×	0	×	0
1	1	1	0	0	×	1	×	1

Maps for combinational circuit



Logic diagram of a 2bit binary counter.



QA: Design a sequential circuit

